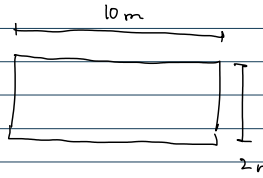
Tuesday, 28 November 2023 10:16 AM

Sliced flat plate:



$Re_L = \frac{\rho V_\infty L}{\mu} = \frac{(1000)(5)(10)}{(0.001)}$

$= 5 \times 10^7$ (turbulent)

For a BL:

Eqn $C_D = \frac{F_D}{(KE)(area)}$ (No laminar flow contr.)

$\frac{0.031}{(Re_L)^{1/7}} = \frac{F_D}{\frac{1}{2}\rho V_\infty^2 \cdot (20)}$

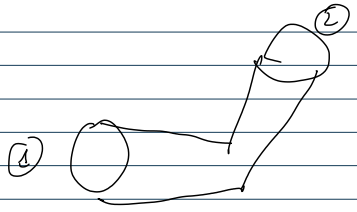
$F_D = \frac{(0.031)((0.5)(1000)\cdot 5^2 \cdot 20)}{(5\times10^7)}$

$F_D = 1.55 \times 10^{-4}$ N

$\times 2$ 3.1×10^{-4}

Atleast ↗

Q2)



Mass balance:

@SS $\rho_1 \langle v_1 \rangle A_1 = \rho_2 \langle v_2 \rangle A_2$; $\rho_1 = \rho_2$

$1(3) = \langle v_2 \rangle 1$

$\langle v_2 \rangle = 3$

MEB: $\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2} + g\Delta z + F = \frac{W_s}{m}$ (0, 0)

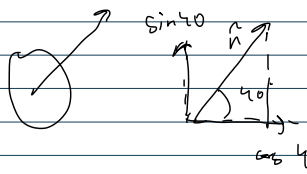
$\frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} = 0$ $\alpha = 1$ (turbulent)

$\frac{P_2 - P_1}{2} = -\left(\frac{3^2 - 1^2}{2}\right)$ 9-1

$1 \times 10^5 - P_1 = -4$

$P_1 = 8 + (1 \times 10^5)$ kg

(c)



$\beta = 1$ (turbulent) (Assuming no friction)

(d) Momentum: $\sum \underline{F} = \rho \cos\theta \frac{(V_1)^2}{\beta} A_1 \cdot \hat{e}_1 + \rho \cos\theta \frac{(V_2)^2}{\beta} A_2 \cdot \hat{e}_2$